Breakthrough Technologies and the Labor Share: Dynamic Evidence from U.S. Sectors

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Abstract

This study aims to estimate the dynamic effect of breakthrough technologies on the U.S. labor share over the period 1987–2025. While existing literature often focuses on the impact of technological innovation on employment, this paper analyzes functional income distribution, asking who captures the gains from increases in productivity due to innovation: capital or labor? We proceed in two stages. First, a descriptive analysis documents the co-movement between the labor share and the adoption of radical technologies. Second, we move to a sector-by-year panel to identify causal effects. We construct a novel treatment variable measuring sectoral *exposure* to a small, time-staggered set of breakthrough technologies, using a shift-share design based on predetermined sector-of-use shares. To capture the gradual diffusion of technology and the path-dependent adjustment of factor shares, we employ Panel Local Projections. This approach allows us to estimate the impulse response function of the labor share directly, providing a rigorous test for pre-trends and avoiding the dynamic misspecification risks of standard panel VARs. Preliminary evidence suggests that the elasticity of the labor share with respect to innovation is not invariant but varies significantly across technological waves, driven by both within-sector displacement and between-sector reallocation.

1 Introduction and Motivation

Recent technological developments have sparked a proliferation of studies on the impact of automation on the labor market. The prevailing literature has largely focused on absolute outcomes: employment levels, wage rates, and skill requirements. Less explored, however, is the relationship between breakthough technological innovation and *functional income distribution*—that is, the division of value added between labor and capital. Analyzing the labor share clarifies how productivity gains are appropriated and reveals the structural margins of inequality inherent in different technological paradigms. Studying the impact of innovation on the labor share is particularly relevant in the present context of increasing inequality: since labor income is generally more evenly distributed across the population than capital income, a shift from labor to capital inevitably widens the income gap.

It is not theoretically settled that automation invariably lowers employment or wages; compensation mechanisms have long been debated (Acemoglu and Restrepo, 2018; Aghion, Antonin,

et al., 2022). Recent work even highlights the job-creating potential of "augmentation innovation" (Autor, Chin, et al., 2024). However, these effects may differ fundamentally when viewed through the lens of factor shares. A technology might preserve employment while drastically reducing the labor share if it increases capital intensity or profit margins disproportionately.

This study hypothesizes that the impact of technological innovation is not invariant across historical phases. Different technological waves—from post-war mechanization to the digital revolution and the recent rise of AI—possess distinct economic properties, diffusion patterns, and institutional contexts. For instance, artificial intelligence presents an extreme case where replication costs are near-zero and training data is often provided by users at no marginal labor cost, potentially altering the profit function in ways distinct from previous mechanical automation.

Furthermore, the impact of innovation is heterogeneous across sectors. Aggregate trends often mask a composition effect: the decline in the aggregate labor share may stem not only from a reduction *within* sectors but also from a reallocation of economic activity towards sectors that are structurally capital-intensive (e.g., platforms, finance) (Montobbio et al., 2022). To disentangle these channels, this paper moves from a macro-aggregate description to a meso-industrial analysis, leveraging sectoral data to isolate industry-specific sources of inequality.

The geographic focus is the United States, chosen for its position at the technological frontier and the availability of consistent long-run data on patents, factor incomes, and industry structure.

2 Related Literature

The decades-long and ongoing decline in the labor share has been explained in the economic literature through various channels. One of the most significant recent contributions on this topic is Bergholt, Furlanetto, and Maffei-Faccioli, 2022. The authors evaluate four potential drivers of the decline in the labor share: (i) rising market power of firms, (ii) higher investment-specific technological growth, (iii) the widespread emergence of automation, and (iv) falling market power of workers. They conclude that the labor income share falls permanently in response to both automation shocks and increases in firms' market power. This finding is particularly relevant for analyzing recent technological innovation, as automation and innovation adoption are predominantly driven by larger firms with substantial market power. This is the case for "superstar firms" described in Autor, Dorn, et al., 2020, that are characterized by low labor share, so that a shock, e.g. technological, that reallocated market share toward these firms, will tend to depress the labor share in the aggregate.

Complementing this aggregate perspective, Autor, Dorn, et al., 2017 provide the foundational empirical evidence detailing the sectoral mechanics behind this reallocation. Analyzing US Economic Census data from 1982 to 2012, they document a pervasive rise in sales concentration–measured by the market share of the top 4 and top 20 firms–across six major sectors, including manufacturing, retail, and services. Crucially, the authors establish that the decline in the ag-

gregate labor share is driven almost entirely by this between-firm reallocation within industries, rather than by a fall in the labor share of the average firm. In fact, the unweighted mean labor share across firms remained relatively stable; the aggregate drop occurred because economic activity shifted toward "superstar" firms that, due to high markups and scalable technologies, inherently operate with lower labor shares. Furthermore, they demonstrate a strong negative cross-industry correlation: sectors experiencing the largest increases in concentration exhibited the steepest declines in labor share, particularly in the post-1990 period. These sectoral findings form the empirical bedrock for the "superstar firm" hypothesis, confirming that technological shocks and changing competitive conditions interact to favor dominant firms, thereby structurally depressing the aggregate labor share through a composition effect.

Complementing the analysis of firm heterogeneity, Mendieta-Muñoz, Rada, and Arnim, 2021 provide a detailed sectoral decomposition of the US labor share from 1948 to 2017, distinguishing between the "golden age" (1948–1979) and the "neoliberal era" (1979–2017). Using a Log Mean Divisia Index (LMDI) approach across 14 sectors, they find that the aggregate decline is driven primarily by the divergence between real compensation and labor productivity growth within sectors, rather than by shifts in employment shares. Specifically, the manufacturing sector plays a critical role: while it buffered the labor share during the golden age through strong wage gains, its post-1979 trajectory is characterized by accelerating productivity growth that vastly outpaces real compensation, coinciding with a sharp contraction in employment. Conversely, the expansion of service sectors—characterized by lower productivity growth but higher labor shares—has acted as a buffer, mitigating what would otherwise be a steeper aggregate decline.

Recent literature has increasingly focused on the institutional origins of technological change, moving beyond the treatment of technology shocks as aggregate primitive disturbances. A pivotal contribution in this strand is provided by Gazzani et al., 2026, who disentangle the macroeconomic effects of innovation by distinguishing between funding sources and ownership regimes. Constructing novel postwar U.S. patent series, the authors identify three distinct categories: privately funded and owned patents, government-funded but privately owned patents, and fully government-owned patents. Using local projections disciplined by external instruments, they demonstrate that government-funded but privately owned innovations–though representing only a small fraction of total filings–are a primary driver of medium-run fluctuations in TFP and GDP, accounting for approximately one-fifth of their forecast error variance. Their findings suggest that the social returns to public R&D are roughly twice those of private R&D, primarily because public funding supports upstream, science-based research (notably by the NIH and NSF) that crowds in subsequent private investment and R&D.

Heer, Irmen, and Süßmuth, 2023 argue that the decline in capital taxation led to an increase in the capital share of national income. In this context, the rising capital share, coupled with a declining workforce in Western economies, created a strong incentive for firms to automate. According to the authors, it is the US tax system itself that generates excessive automation incentives, resulting in a sub-optimally low labor share. Other causes are analyzed by the very

recent study of Budrys, Porqueddu, and Sokol, 2026 who investigate the macroeconomic effects of wage bargaining shocks in Germany by employing a Structural Vector Autoregression (SVAR) model identified through a novel combination of sign and narrative restrictions. Unlike traditional approaches that rely solely on theoretical sign constraints, the authors exploit exogenous variations in bargaining power derived from specific historical events: the introduction of the statutory minimum wage in 2015 and major industrial disputes led by the IG Metall union between 1995 and 2018. This narrative identification strategy sharpens inference on the dynamic responses of unemployment and output, which are left unrestricted a priori. The study finds that positive wage bargaining shocks lead to a persistent increase in unemployment and a temporary decline in output, consistent with search-and-matching models featuring rigid labor markets. Furthermore, the authors document a near-complete pass-through of wage increases to consumer prices and show that while such shocks initially raise the labor share, this effect reverses in the medium term due to sustained employment losses. The results underscore the significance of labor market institutions and bargaining dynamics as key drivers of business cycle fluctuations in inflation and employment.

Not so different results are those of González, 2025 that, using a structural vector autoregression (SVAR) identified by narrative sign restrictions, isolates exogenous changes in workers' bargaining power based on three pivotal U.S. institutional events: the 1952 steel seizure under Truman, the 1962 executive order granting collective bargaining rights to federal employees under Kennedy, and the 1981 PATCO strike decertification under Reagan. The study finds that positive bargaining shocks unambiguously reduce output by 0.4–0.8 percent in the long run while having negligible or negative effects on the labor share, as the resulting economic contraction creates labor market slack that moderates wage demands. Crucially, the author demonstrates that demand shocks, rather than bargaining shocks, account for approximately three-quarters of the variance in the labor share at long horizons. In a counterfactual analysis, González shows that absent the decline in bargaining power between 1997 and 2012, the recent fall in the labor share would have been even more pronounced, suggesting that weak aggregate demand is the dominant factor behind the functional distribution of income trends in recent decades.

Minniti, Prettner, and Venturini, 2025 find that AI innovation reduces the labor share mainly through wage compression, particularly for high- and medium-skill workers whose employment shares remain stable. In contrast, for low-skill workers, a mild expansion in employment partially offsets the wage decline. However, it is important to note that they proxy AI innovation using patent data, which generally captures innovation production rather than actual adoption. Furthermore, their study only covers data until 2017, thus missing LLM. Aghion, Akcigit, et al., 2019 highlight that as innovation increases particularly through patenting, the income share accounted for by non entrepreneurial wage income tends to decline, highlighting the unequal distribution of innovation rents. Regarding the heterogeneity in the impact of technological shocks, Bertinelli, Cardi, and Restout, 2022 demonstrate that innovation effects are not uniform between traded and non-traded sectors. Specifically, they find that such shocks induce a labor reallocation toward the non-traded sector, accounting for 35% of the rise in non-traded hours

worked. However, their analysis relies on a reductive conceptualization of technological innovation, defined simply as shocks to Total Factor Productivity (TFP). The authors empirically estimate the elasticity of substitution between capital and labor (σ) to be less than one (0.69 for the traded sector and 0.72 for the non-traded sector), implying that capital and labor are gross complements. This parameter is critical to their analysis, as it necessitates the introduction of labor-biased technological change to explain the rising labor share amidst falling capital-labor ratios.[1]

Challenging the consensus on a global decline, Gutiérrez and Piton, 2020 argue that cross-country comparisons of corporate labor shares are severely biased by differences in national accounting standards, specifically regarding the treatment of housing and self-employment. While the US corporate sector excludes these components, many European and advanced economies include significant amounts of residential real estate and self-employed workers in their corporate definitions, artificially depressing measured labor shares as these components have grown. By constructing "harmonized" non-housing labor share series for major advanced economies from 1970 to 2015, the authors show that the labor share has remained stable or even increased in most countries outside the US and Canada. The apparent global decline is thus largely a measurement artifact driven by the US experience and inconsistent sectoral delineations. These findings cast doubt on technological explanations—such as automation or falling capital prices—which predict uniform global trends, and instead point to country-specific institutional factors, measurement issues, or declining competition in select US industries as the primary drivers of the observed divergence.

Focusing on the US institutional context, Smith et al., 2022 demonstrate that a significant portion of the observed decline in the corporate labor share is attributable to the rise of pass-through business entities (S-corporations and partnerships) following the Tax Reform Act of 1986. As firms reorganized to minimize tax liabilities, owner-managers increasingly recharacterized labor income as profit income, particularly within S-corporations. Furthermore, labor-intensive service activities shifted from the corporate sector to the partnership sector. The authors estimate that adjusting for these organizational changes—by reclassifying recharacterized wages and reincorporating partnership activity—reduces the measured decline in the US corporate labor share by approximately 30%. Their analysis reveals that the decline is almost exclusively a manufacturing phenomenon, which was fully offset by the rise of the service sector until 2000 and only partially offset thereafter. This suggests that tax-induced organizational shifts, rather than a broad-based technological displacement of labor, explain a substantial fraction of the aggregate trend.

Karabarbounis and Neiman, 2014 document a significant global decline in the labor share

[1]It should be noted that the very estimation of such an aggregate substitution coefficient presupposes that capital can be treated as a single, homogeneous aggregate, an assumption that has been extensively criticized and disputed in the literature. This critique, most prominently associated with Joan Robinson and the broader Cambridge Capital Controversy (involving economists such as Piero Sraffa and Luigi Pasinetti), argues that capital cannot be meaningfully aggregated independent of distribution and prices, thereby challenging the theoretical coherence of the aggregate production function and the stability of the estimated elasticity.

of income since the early 1980s, affecting the majority of countries and industries. The authors attribute approximately half of this 5-percentage-point drop to the falling relative price of investment goods, driven by advances in information technology, which incentivized firms to substitute labor with capital. Crucially, the study finds that this trend stems primarily from within-industry changes rather than shifts in economic composition, and it distinguishes the welfare implications of this shift: unlike a labor share decline caused by rising profit markups, the substitution driven by cheaper investment goods is associated with overall gains in output and economic welfare. Note that here, instead, they estimated a coefficient of substitution between capital and labor of about 1,25.

A recent contribution by Bridgman and Herrendorf, 2022 finds that increasing markups account for the majority of the decline in the labor share. This effect is significantly amplified by input-output linkages incorporated into their multi-sector model, which generate double marginalization. Indeed, although capital deepening exerted large direct effects in the goods sector, structural change failed to offset this decline because markups rose across both sectors, rendering the reallocation effect insufficient to counteract the fall in the labor share. Furthermore, the authors provide a robust framework to isolate pure profits (factorless income) from the return to capital. This distinction is crucial, since as mentioned the debate on the dynamics of labor and capital shares is often complicated by challenges in decomposing the capital share.

Regarding the economic effects of innovation, Cascaldi-Garcia, Vukotić, and Zubairy, 2023 provide critical insights into the state-dependent nature of technological shocks. They demonstrate that aggregate patent-based innovation shocks exert a more pronounced impact on the economy during recessions than in expansions, a phenomenon driven primarily by a significant surge in private investment. These findings lend empirical support to the Schumpeterian hypothesis that recessions create opportune conditions for productivity-enhancing activities, as the lower opportunity cost of investment during downturns may outweigh binding financial constraints. Crucially, the authors highlight that this response is heterogeneous: firms with stronger financial health (low default risk) or those constrained by equity issuance are able to capitalize on these opportunities, whereas debt-constrained firms remain limited. Methodologically, they identify these innovation shocks by exploiting changes in the stock market valuations of firms upon patent grants, thereby extending the foundational approach of Kogan et al., 2017 to explicitly analyze business cycle dynamics and the role of financial frictions.

Of the most crucial relevance in this literature remains the problem of how technological shocks are identified. A significant contribution to the recent literature on the identification of technology shocks is provided by Miranda-Agrippino, Hacıoğlu-Hoke, and Bluwstein, 2026, who propose a novel empirical strategy to isolate *technology news shocks*. The authors construct an external instrumental variable based on the flow of patent applications filed with the USPTO, purged of pre-existing macroeconomic expectations and contemporaneous fiscal and monetary policy shocks. Employing this instrument within an SVAR-IV framework, Miranda-Agrippino, Hacıoğlu-Hoke, and Bluwstein demonstrate that positive news about future techno-

logical progress triggers a broad-based and anticipatory economic expansion–characterized by rises in output, consumption, investment, and hours worked–well before any material improvement in Total Factor Productivity (TFP) is recorded. Their findings indicate that while such shocks are a fundamental driver of long-run growth and the trend component of TFP, they explain only a modest fraction of macroeconomic volatility at business cycle frequencies.

Still, the present study diverges conceptually from the in the nature and transmission of these shocks. Approaches like Miranda-Agrippino, Hacıoğlu-Hoke, and Bluwstein, 2026 identify innovation through the lens of financial markets, treating the grant of a patent as a discrete *news shock* where the immediate stock market reaction proxies the expected future value of the technology. In this framework, the shock is instantaneous and driven by investor expectations. In contrast, this thesis adopts a framework of *structural exposure shocks* rooted in evolutionary theories of technological diffusion. Here, the shock is not merely the announcement of an invention, but the gradual wave of adoption resulting from the interaction between the global supply of breakthrough technologies and a sector's predetermined structural propensity to utilize them (the Shift-Share design). This distinction is critical for analyzing the labor share: whereas news shocks capture financial anticipations that may decouple from real implementation, our approach isolates the *real* pressure of technology adoption as it physically penetrates production processes over time. By modeling diffusion explicitly rather than relying on market sentiment, we avoid conflating stock market optimism with the actual reorganization of production factors that drives changes in functional income distribution.

Additionally, many of the mentioned studies build the model dynamics on a General Equilibrium framework, as in neoclassical growth models. We deem this to be a limitation of the aforementioned analyses in the context of rapid technological transformation. Structural approaches necessarily rely on strong a priori assumptions regarding the production function (e.g., constant elasticity of substitution), factor mobility, and optimization behaviors that may not hold during discontinuous technological shifts. In contrast, our empirical strategy adopts a reduced-form approach that remains agnostic to specific theoretical structures. By employing Panel Local Projections combined with a Shift-Share design, we do not impose a predefined production function or optimization path. Instead, we let the data reveal the dynamic impulse response of the labor share to technological shocks directly. This flexibility allows us to capture non-linear adjustments, path-dependent diffusion patterns, and heterogeneous effects across sectors that rigid structural models might obscure or mis-specify.

3 Research Questions

The primary objective of this thesis is to analyze the relation between technological innovation and functional income distribution. Specifically, we address three questions:

1. **Heterogeneity across waves:** Do different technological epochs exhibit distinct elasticities of the labor share with respect to innovation?

2. **Causal Impact:** What is the dynamic causal effect of adopting breakthrough technologies on the sectoral labor share? Is the effect immediate, delayed, transitory, or permanent?

3. **Mechanisms:** To what extent is the aggregate decline in the labor share driven by *within-sector* technological displacement versus *between-sector* structural reallocation?

4 Data and Measurement

4.1 Innovation Measures

We measure technological innovation using three complementary indices derived from the universe of U.S. patents:

- The text-based *breakthrough index* measuring technological radicalness (Kelly et al., 2021).
- The *labor-saving index* capturing the displacement potential of patents (Montobbio et al., 2023).
- The stock-market-based *technological value* measure (Kogan et al., 2017), extended by Cascaldi-Garcia, Vukotić, and Zubairy, 2023.

We focus primarily on the breakthrough index as our baseline treatment, using the labor-saving index for robustness. Given the high correlation between radicalness and labor-displacement in the data, including both simultaneously induces multicollinearity; instead, we verify that results hold when swapping the definitions of "breakthrough."

4.2 Selection of Breakthrough Technologies

To operationalize the treatment variable, we identify a discrete set of ten breakthrough technological waves spanning the period 1970–2025. This selection is not arbitrary but is driven by the joint distribution of our innovation metrics: we target technologies that exhibit simultaneous peaks in the text-based *breakthrough index* (indicating high scientific novelty and forward impact à la Kelly et al.) and the stock-market-based *technological value* index (indicating high expected private rents à la Kogan et al.). By intersecting these two dimensions, we isolate innovations that represent both a paradigm shift in the knowledge frontier and a substantial reallocation of market value. The selected technologies, staggered in time, are expected to be alligned to the following technological waves:

1. **Microelectronics and VLSI Circuits (1970–1978):** Characterized by the transition to microprocessors and high-density memory, marking the onset of hardware-based automation in manufacturing.

2. **Personal Computing and Graphical Interfaces (1979–1988):** Encompassing patents related to the PC architecture, mouse, and GUI, driving the computerization of administrative and routine cognitive tasks.

3. **Recombinant Biotechnology (1980–1990)**[2]

4. **Internet Infrastructure and Networking Protocols (1990–1998):** Covering packet-switching, routing algorithms, and browser technologies, which drastically reduced communication costs and enabled the first wave of service offshoring.

5. **Mobile Telecommunications and Wireless Standards (1995–2005):** Spanning GSM/CDMA standards and early smartphone architectures, optimizing logistics and displacing intermediate information services.

6. **Search Algorithms and Digital Platforms (1998–2008):** Including PageRank, recommendation engines, and secure e-commerce protocols, facilitating disintermediation in retail and media sectors.

7. **Cloud Computing and Virtualization (2005–2015):** Defined by server virtualization and distributed architectures, leading to the consolidation of IT infrastructure and a reduction in systemic maintenance labor.

8. **Renewable Energy and Advanced Storage (2008–2020):** Covering high-efficiency photovoltaics, wind turbine optimization, and Li-ion battery management systems, driving capital deepening in the energy sector.

9. **Sharing Economy and Service Automation (2010–2020):** Encompassing real-time matching algorithms and mobile payment systems, transforming labor organization in transport and hospitality via the "gig" model.

10. **Generative AI and Deep Learning (2015–2025):** The most recent wave, including transformer architectures and large language models, targeting non-routine cognitive tasks and potentially decoupling productivity growth from labor demand in high-skill sectors.

This stratified selection allows us to test for heterogeneity in the labor share response across distinct technological paradigms. Specifically, it enables us to distinguish between early waves that primarily substituted routine manual and cognitive labor (Items 1–2) and recent waves (Items 9–10) that may substitute non-routine tasks and alter the bargaining power of high-skill workers. For each technology $\tau \in \mathcal{T}$, we construct a specific exposure series using Eq. (1), allowing the Panel Local Projections to estimate distinct impulse response functions for each technological epoch.

[2]The inclusion of recombinant biotechnology (1980–1990) warrants specific clarification regarding its transmission mechanism to the labor share, as it differs fundamentally from routine-biased automation. While this technological wave is characterized by extreme *skill-biased technical change*—substituting traditional chemical processes with high-wage scientific labor in the R&D phase—its net effect on the functional income distribution is hypothesized to be negative. This occurs through two primary channels: (i) scalability, where negligible marginal costs decouple productivity from employment; and (ii) rent appropriation via patents, shifting income to capital. Thus, while average wages rise, the labor share falls in favor of profits and royalties, highlighting the distinction between wage levels and factor shares.

4.3 Sectoral Exposure Construction

A key challenge is mapping patent-level data to sectoral outcomes. We adopt a *sector-of-use* logic: a sector is credited with the technologies it *employs* in production, not only those it produces. This is crucial for service and digital sectors that adopt heavily but patent little.

Let $\mathcal{T} = \{\tau_1, \ldots, \tau_K\}$ be a selected set of breakthrough technologies. The exposure of sector s in year t is constructed as:

$$\text{Exposure}_{s,t} = \sum_{\tau \in \mathcal{T}} \pi_{s\tau} \cdot \left[\sum_{p \in \tau} B_p \, D_{s\tau}(t - t_p) \right], \tag{1}$$

where:

- $\pi_{s\tau}$ are predetermined *use shares* derived from the Algorithmic Links with Probabilities (ALP) concordance (Lybbert and Zolas, 2014), fixed at a pre-sample base year to ensure exogeneity.
- B_p is the breakthrough score of patent p.
- $D_{s\tau}(\cdot)$ is a diffusion function (e.g., logistic S-curve) mapping the patent filing date t_p to the adoption path in sector s.

Timing of the Technological Shock: The Primacy of Application Dates

A critical methodological choice in our design concerns the temporal anchoring of the technological shock. We define the onset of the shock ($t = 0$) as the patent *application* date (t_p), rather than the grant date. This choice is grounded in recent empirical evidence on the transmission of technological information. Miranda-Agrippino, Hacıoğlu-Hoke, and Bluwstein, 2026 demonstrate that patent applications contain significant informational content that becomes public immediately upon filing, triggering measurable macroeconomic responses (e.g., in investment and asset prices) well before patents are officially granted or technologies are fully diffused. They show that using grant dates introduces a substantial lag that conflates the arrival of news with administrative processing times, thereby obscuring the anticipatory behavior of economic agents.

By anchoring our exposure measure to the application date, we align our identification strategy with the moment the technological opportunity set expands for the economy. While our diffusion function $D(t - t_p)$ models the *gradual physical adoption* of the technology (which necessarily takes time), starting the clock at the application date ensures that our impulse response functions capture both the immediate *expectation-driven* adjustments (e.g., hiring freezes or restructuring announcements made in anticipation of the technology) and the subsequent *implementation-driven* effects. This approach mitigates the risk of *look-ahead bias* inherent in grant-based measures and allows the Local Projections framework to empirically disentangle the timing of financial expectations from the pace of real economic transformation.

Modeling the Diffusion Path: Accounting for Expectations and Adoption

A critical component of our exposure measure is the diffusion function $D_{s\tau}(t - t_p)$, which maps the time elapsed since the patent application into the intensity of technological exposure for sector s. Standard approaches in the literature often assume a gradual S-shaped diffusion (e.g., logistic or Gompertz curves), reflecting the slow physical embodiment of new technologies in capital stock and production processes. However, the aforementioned study by Miranda-Agrippino, Hacıoğlu-Hoke, and Bluwstein, 2026 also highlights that the economic impact of innovation begins immediately upon the filing of a patent application, as agents update their expectations about future productivity and profitability well before the technology is physically deployed.

To reconcile these two dimensions—the instantaneous *news effect* and the gradual *adoption effect*—our baseline specification allows for a *modified diffusion profile*. Rather than imposing a function that starts strictly at zero, we consider functional forms that permit an initial positive weight at $t = t_p$ (capturing the immediate anticipatory adjustment of firms, such as hiring freezes or strategic reallocations driven by expectations), followed by a monotonic increase reflecting the physical diffusion of the technology. Formally, this implies a structure where $D(0) \geq 0$, with the magnitude of the initial jump governed by a parameter $\alpha \in [0, 1]$ that weighs the relative importance of expectation-driven channels versus implementation-driven channels.

In this draft, we remain agnostic regarding the precise functional form and the exact calibration of α. We will empirically test alternative specifications: (i) a pure adoption model where $D(0) = 0$ (standard S-curve); (ii) a hybrid model with a conservative initial jump (e.g., $D(0) > 0$) consistent with the news-shock literature; and (iii) a fully data-driven approach where the shape of the effective diffusion is revealed by the sequence of impulse responses $\{\beta_h\}$ in the Local Projections. This flexibility ensures that our estimates are not biased by a priori assumptions on the speed of adjustment, allowing the data to determine whether the labor share responds primarily to the *announcement* of breakthrough technologies or to their *material deployment*.

4.4 Labor Share Data

Factor shares are drawn from official U.S. statistics:

- **Aggregate Level (1970–2025):** Labor share from BLS major-sector productivity series; profit share from BEA NIPA tables.
- **Sectoral Panel (1985–2025):** Industry-level data from the BEA–BLS KLEMS Production Accounts (NAICS classification).

For the recent period (post-1995), we complement patent data with intangible investment measures from EUKLEMS & INTANProd to capture non-patentable innovation (software, organizational capital) prevalent in the digital economy.

Sample Period and Data Consistency

The baseline sectoral panel analysis covers the period **1987–2025**. This start date is dictated by the need for strict data comparability rather than mere availability. The year 1987 marks the official transition in the U.S. from the *Standard Industrial Classification* (SIC) to the *North American Industry Classification System* (NAICS). While aggregate series exist prior to this date, sectoral data at the granular level required for our Shift-Share design (2- to 3-digit detail) suffers from structural breaks and conceptual mismatches when splicing SIC and NAICS definitions.

By restricting the causal analysis to the post-1987 NAICS era, we ensure that: (i) the sectoral boundaries remain conceptually consistent over time, particularly for the service and technology sectors central to our study; (ii) the mapping between patent classes and industries via the ALP concordance is direct and free from conversion noise; and (iii) the factor share data from BEA KLEMS is fully harmonized. This choice prioritizes measurement accuracy and identification validity over sample length, ensuring that the estimated dynamics reflect genuine technological shocks rather than artifacts of industrial reclassification.

Data Sources and Availability

The empirical analysis relies on the integration of three distinct data sources covering the period 1970–2025 at varying levels of granularity.

- **Patent Data and Innovation Indices:** We utilize the full universe of U.S. patents granted by the USPTO. Patent text and citation data are sourced from the USPTO Bulk Data repository and the NBER Patent Data Project (for historical coverage up to 2010). The breakthrough and labor-saving indices are constructed following the methodologies of Kelly et al. (2021), Kogan et al. (2017), and Montobbio et al. (2023), leveraging text-based NLP techniques and stock market reaction data. Where necessary to extend coverage beyond the original sample end-dates of these studies, we update the indices using recent USPTO filings and citation windows.

- **Technology-Industry Mapping:** The sector-of-use shares ($\pi_{s\tau}$) are derived from the *Algorithmic Links with Probabilities* (ALP) concordance developed by Lybbert and Zolas (2014). This publicly available dataset links International Patent Classification (IPC) codes to NAICS industries based on semantic similarity between patent texts and industry descriptions. We fix these shares to a pre-sample base year to ensure exogeneity.

- **Sectoral Economic Accounts:** Industry-level labor shares, value added, and compensation data are drawn from the BEA–BLS Industry-Level Production Account (KLEMS) for the period 1987–2025, which provides consistent NAICS-based series. For the longer historical aggregate analysis (1945–1986), we rely on the BLS Major Sector Productivity series and BEA NIPA tables.

- **Intangible Investment (Robustness):** To account for non-patented innovation in services, we incorporate data on intangible capital formation (software, R&D, organizational

capital) from the EUKLEMS & INTANProd database, available for the U.S. from 1995 onwards.

The convergence of these sources allows for the construction of a novel panel dataset where time-varying technological shocks are interacted with structural sectoral characteristics, covering the most critical decades of technological transformation in the U.S. economy.

Robustness Check: Long-Run Analysis with Aggregated Sectors

To verify that our baseline results are not driven by specific conditions of the post-1987 period (e.g., the IT boom or the rise of platforms), we conduct a robustness exercise extending the sample back to **1970**.

Given the structural break between SIC and NAICS classifications, this extension requires aggregating the data to a higher level of granularity (**1-digit major divisions and consistent 2-digit major groups**). We construct a balanced long-run panel by: (i) aggregating the post-1987 NAICS data from BEA KLEMS to these broad categories; (ii) mapping pre-1987 SIC data from BLS and Census sources to the same categories using official concordance tables; and (iii) aggregating the ALP sector-of-use shares ($\pi_{s\tau}$) to match this broader industrial definition.

While this higher level of aggregation reduces cross-sectional variation and may attenuate the precision of the Shift-Share identification, it serves as a critical validity test. If the negative impulse response of the labor share to breakthrough exposure persists in this long-run, coarser panel, it confirms that our findings are not an artifact of the NAICS era or recent technological waves, but reflect a structural relationship spanning five decades of U.S. industrial history.

Crucially, we estimate the same Panel Local Projections specification on this long-run aggregated panel. This allows us to compare the dynamic impulse response functions (β_h) across the two samples. While the reduced cross-sectional variation in the 1-digit/2-digit panel may lead to wider confidence intervals, observing a qualitatively similar dynamic profile (e.g., timing and persistence of the labor share decline) would provide strong evidence that our baseline results are not driven by the specific granularities of the NAICS era but reflect a persistent structural dynamic of technological adoption.

5 Empirical Strategy

5.1 Identification Strategy: A Shift-Share Approach

Identification rests on a shift-share (Bartik) design tailored to the study of a small, selected set of breakthrough technologies. Since we do not rely on the law of large numbers over many independent shocks, identification exploits cross-sectional variation in *exposure* driven by predetermined sectoral structures ($\pi_{s\tau}$). The core identifying assumption is that a sector's historical propensity to use specific technologies—its inherited technological "DNA"—is uncorrelated with future idiosyncratic shocks to its labor share, conditional on fixed effects. This approach treats the diffusion of breakthrough technologies as an external wave that hits sectors differently based on their pre-existing structural sensitivity.

5.2 Panel Local Projections

To estimate the *dynamic* response of the labor share to technology adoption, we employ **Panel Local Projections** (LP) (Jordà, 2005). Unlike Vector Autoregressions (VARs), which impose a rigid dynamic structure that may be misspecified for slow-diffusing, path-dependent technologies, Local Projections estimate the impulse response function directly at each horizon h. This flexibility is crucial for capturing the non-linear adjustment processes typical of evolutionary technological change.

For each horizon $h \in \{-L, \ldots, H\}$, we estimate the following regression:

$$\ln LS_{s,t+h} - \ln LS_{s,t-1} \;=\; \alpha_{s,h} + \delta_{t,h} + \beta_h \,\text{Exposure}_{s,t} + \gamma_h' X_{s,t} + \varepsilon_{s,t+h}, \tag{2}$$

where:

- The dependent variable is the cumulative change in the labor share from $t-1$ to $t+h$.
- $\text{Exposure}_{s,t}$ is the exposure measure defined in Eq. (1), representing the sector-specific intensity of the technological wave.
- $\alpha_{s,h}$ and $\delta_{t,h}$ are sector and year fixed effects specific to horizon h, absorbing time-invariant heterogeneity and common macro shocks.
- $X_{s,t}$ includes time-varying sectoral controls measured at time t, representing the state of the sectoral selection environment (discussed below).
- β_h traces the impulse response of the labor share to a unit shock in breakthrough exposure.

Advantages of this Approach

1. **Flexible Dynamics:** The sequence $\{\beta_h\}_{h\geq 0}$ reveals whether the labor share adjustment is immediate or delayed, and whether effects are permanent or transitory, without imposing restrictive autoregressive parameters.
2. **Pre-Trend Validation:** Estimating Eq. (2) for leads ($h < 0$) provides a direct falsification test. If $\beta_h \approx 0$ for $h < 0$, it confirms that exposed sectors were not on diverging trends prior to adoption, supporting the exogeneity of the exposure measure.
3. **Robustness to Misspecification:** LPs are more robust to non-linearities and structural breaks than standard panel VARs, which is critical given the long sample period spanning distinct technological regimes.

Standard errors are clustered at the sector level to account for serial correlation. To reinforce causal identification and align with the systemic nature of innovation diffusion, we complement the baseline OLS estimates with a **Local Projection IV** (LP-IV) specification. In this approach, the exposure variable $\text{Exposure}_{s,t}$ is instrumented using a *leave-one-out* shift-share construct. This instrument captures the *systemic technological pressure* exerted on sector s by the rest of the economy, leveraging the sector's path-dependent structural sensitivity while excluding its own endogenous feedback on technology diffusion. Comparing the OLS and IV

impulse responses serves as a test for endogeneity bias. If the IV estimates (which isolate the exogenous supply push of technology) are consistent with the OLS estimates, it suggests that the baseline results are not driven by reverse causality or omitted time-varying confounders correlated with our instrument. Conversely, a significant divergence would indicate that sectoral labor share trends themselves influence technology adoption, necessitating the IV approach for causal inference.

Control Variables and Instrumental Variable Strategy in an Evolutionary Perspective

To ensure causal identification while respecting the evolutionary nature of sectoral dynamics, our specification includes a vector of time-varying sectoral controls ($X_{s,t}$) measured at the time of the shock (t). In an evolutionary framework, these are not merely statistical adjustments to hold factors constant; they represent the *structural state variables* of the sectoral *selection environment*. They define the context in which new technologies are selected and absorbed. Specifically, we include:

1. **Capital-Labor Ratio** (from BEA KLEMS): Accounts for the pre-existing technological regime and path-dependent capital deepening, determining the sector's absorptive capacity for capital-embodied innovations.
2. **Skill Composition of Employment** (share of high-skilled workers from CPS/BLS): Proxies for the sector's *related variety* and knowledge base, which condition the ability to adopt complex breakthrough technologies.
3. **Import Penetration Rates** (from BEA Industry Accounts): Captures competitive pressure from global technological frontiers, acting as an external selection force.
4. **Industry Unionization Rates** (from CPS/BLS): Reflects the institutional framework governing factor distribution and the power to negotiate the gains from innovation.
5. **Market Concentration Index** (HHI from Census data): Controls for shifts in market power and rent appropriation, which influence how productivity gains are distributed between capital and labor.

By conditioning on these factors, we acknowledge that the impact of innovation is mediated by the sector's existing factor endowments and institutional context.

Regarding endogeneity, our *Local Projection IV* approach employs a *leave-one-out* instrument ($Z_{s,t}$) constructed to align with the systemic view of innovation. For each sector s, we calculate the global diffusion of breakthrough technologies weighted by s's historical structural shares ($\pi_{s\tau}$), but *excluding* sector s's own contribution to the technology's adoption. This instrument isolates the *exogenous supply push* of technologies available to a sector due to its historical specialization, purging any feedback effect from the sector's own short-term strategic decisions or contemporaneous labor market shocks. It effectively measures how the *rest of the system's* technological evolution drives transformation in sector s, distinct from the sector's internal adaptive dynamics.

5.2.1 Heterogeneity Analysis via Interaction Terms

To investigate whether the impact of breakthrough technologies varies across industrial structures without sacrificing statistical power by splitting the sample, we employ an interaction approach within the full panel regression. Rather than estimating separate models for subsamples (e.g., manufacturing only vs. services only), we augment Equation (2) with interaction terms between the exposure variable and baseline sector-specific characteristics. This allows us to recover distinct impulse response functions for different groups simultaneously while leveraging the full cross-sectional variation of the dataset to estimate fixed effects.

Specifically, we estimate the following specification for each horizon h:

$$\Delta \ln LS_{s,t+h} = \sum_{g \in G} \beta_{h,g} \left(Exposure_{s,t} \times D_{s,g} \right) + \alpha_{s,h} + \delta_{t,h} + \gamma_h' X_{s,t} + \varepsilon_{s,t+h}, \tag{3}$$

where $D_{s,g}$ is a dummy variable indicating membership in a specific sectoral group g (e.g., $D_{s,\text{Manuf}} = 1$ if sector s belongs to Manufacturing, and 0 otherwise), or a standardized continuous measure of a structural characteristic (e.g., initial capital intensity or skill composition). The coefficient $\beta_{h,g}$ traces the impulse response of the labor share to a unit shock in breakthrough exposure specifically for group g.

This specification offers three key advantages:

1. **Efficiency:** By pooling all sectors in a single estimation, we maximize the degrees of freedom available for estimating the sector ($\alpha_{s,h}$) and time ($\delta_{t,h}$) fixed effects, leading to more precise standard errors compared to split-sample regressions.

2. **Direct Testing:** It enables formal statistical testing of heterogeneity. We can directly test the null hypothesis that the technological shock affects all sectors equally ($H_0 : \beta_{h,\text{Manuf}} = \beta_{h,\text{Serv}}$) using standard Wald tests on the interaction coefficients.

3. **Granularity:** We can explore multiple dimensions of heterogeneity simultaneously, such as distinguishing between traded and non-traded sectors, or analyzing quartiles of pre-sample labor share, to identify which structural features amplify or mitigate the distributional effects of innovation.

In the results section, we will present the estimated impulse responses $\beta_{h,g}$ for key groupings to disentangle whether the aggregate decline in the labor share is driven by pervasive automation across all industries or concentrated reallocation within specific technological adopters.

5.2.2 Alternative interaction term for heterogeneity analysis

To explore the heterogeneity of technological impacts at the highest level of granularity without sacrificing statistical power, we employ an *imputed response* approach. Rather than estimating separate regressions for each of the N sectors—which would suffer from low degrees of freedom and contaminated control groups—we estimate a single unified model that allows the impulse response to vary continuously with sector-specific characteristics. We augment our baseline

Local Projection specification with an interaction between the exposure variable and a time-invariant sectoral characteristic Z_s (e.g., the sector's long-run average exposure intensity, or baseline capital-labor ratio):

$$\Delta \ln LS_{s,t+h} = \beta_{h,0} Exposure_{s,t} + \beta_{h,1}(Exposure_{s,t} \times Z_s) + \alpha_{s,h} + \delta_{t,h} + \varepsilon_{s,t+h}, \tag{4}$$

where $\beta_{h,0}$ captures the average response for a sector with $Z_s = 0$ (or mean-centered), and $\beta_{h,1}$ measures how the response scales with the characteristic Z_s. From the estimated coefficients $\hat{\beta}_{h,0}$ and $\hat{\beta}_{h,1}$, we construct the *imputed* impulse response function for each individual sector s as:

$$\widehat{IRF}_{s,h} = \hat{\beta}_{h,0} + \hat{\beta}_{h,1} \cdot Z_s. \tag{5}$$

This approach yields a distinct dynamic trajectory for every sector in our sample (e.g., Motor Vehicles, Financial Services, Retail Trade), reflecting its unique structural position (Z_s) within the technological landscape. It combines the statistical efficiency of a pooled panel estimator with the descriptive richness of sector-level analysis.

In the results section, we visualize these heterogeneous dynamics by plotting the family of impulse responses $\{\widehat{IRF}_{s,h}\}_{s=1}^{S}$ for key industries. This allows us to:

- Identify *outlier sectors* that deviate significantly from the average trend due to their extreme structural characteristics.
- Observe the *continuum of effects*: rather than a binary distinction (e.g., Manufacturing vs. Services), we reveal a gradient of responses where sectors with higher Z_s experience systematically deeper (or shallower) declines in the labor share.
- Maintain *theoretical consistency*: all sector-specific estimates are derived from a single coherent structural relationship, avoiding the noise and inconsistency of N separate regressions.

This method aligns with our evolutionary perspective by acknowledging that each sector follows a unique trajectory determined by its specific "technological DNA" (Z_s), while recognizing that these trajectories are governed by common underlying economic mechanisms (β_h).

5.2.3 Selection of Structural Moderators (Z_s)

To uncover the structural drivers of heterogeneity, we interact the breakthrough exposure variable with a set of time-invariant sectoral characteristics (Z_s), measured at the baseline year (1987) to ensure exogeneity. The selection of these moderators is guided by competing theoretical hypotheses on the transmission mechanisms of technological shocks:

1. **Institutional Power (Union Density):** Defined as the share of unionized workers in sector s in 1987. From an institutionalist and post-Keynesian perspective, bargaining power acts as a critical filter determining whether productivity gains are appropriated by labor or capital. We hypothesize that high baseline unionization mitigates the negative impact of automation on the labor share, reflecting the capacity of organized labor to negotiate the fruits of innovation.

2. **Technical Composition (Capital Intensity):** Measured as the real capital-to-labor ratio (K/L) in the baseline period. This variable tests the *substitution hypothesis*: sectors with higher initial capital intensity may possess a productive structure more complementary to radical automation, potentially accelerating the displacement of labor. Conversely, following the critique of aggregate production functions (Cambridge Capital Controversy), this variable also captures the *historical legacy* of investment patterns that predate the current technological wave.

3. **Market Structure (Concentration):** Captured by the Herfindahl-Hirschman Index (HHI) or the concentration ratio of the top four firms ($CR4$) in 1987. Drawing on the "superstar firm" literature (e.g., Autor, Dorn, et al. (2020); Bergholt, Furlanetto, and Maffei-Faccioli (2022)), we investigate whether sectors with pre-existing high market power are more likely to use breakthrough technologies to expand profit margins (markups) rather than increase compensation, thereby driving a wedge between productivity and wages.

4. **Structural Exposure (Average Technological Affinity):** Defined as the long-run average of the sector's exposure score ($\bar{Exposure}_s$) over the pre-sample period. This serves as a baseline test for *dose-response* linearity: do sectors that are structurally more "immersed" in breakthrough technologies exhibit proportionally larger distributional shifts? This moderator isolates the sheer magnitude of the technological pressure from specific institutional or technical mediators.

By estimating equation (4) with each of these moderators in turn, we decompose the average effect into distinct structural channels. This approach allows us to distinguish whether the decline in the labor share is a *technological necessity* (driven primarily by capital intensity or exposure magnitude) or an *institutional contingency* (driven by weak bargaining power and high market concentration). Such differentiation is pivotal for designing targeted policy responses that address the root causes of inequality rather than its technological symptoms.

5.2.4 Addressing Methodological Critiques: Exogeneity and Structural Dynamics:

A potential critique of this approach is that shift-share designs rely on the exogeneity of sectoral shocks, an assumption often associated with partial equilibrium analysis where sectors are treated in isolation. We explicitly reject this interpretation. Our identification strategy does not assume that technology is exogenous in an absolute sense—we acknowledge that innovation is endogenous to the broader economic system, driven by investment, policy, and power dynamics (as discussed in Section 2). Instead, our assumption is more nuanced: conditional on the *historical* structure of sectoral technologies (the fixed shares $\pi_{s\tau}$), the *timing* and *direction* of new global technological waves are external to any single sector's short-term labor market decisions.

We are not isolating a sector from the rest of the economy (partial equilibrium); rather, we are tracing how *structurally differentiated* sectors navigate a common systemic transformation. The variation we exploit is not a vacuum-sealed shock, but the differential pressure exerted by the system on sectors with different inherited "technological DNA." Our estimates therefore

capture the *realized* distributional consequences of this pressure, inclusive of all institutional feedbacks, price adjustments, and demand effects that occur in historical time, without imposing the theoretical constraint that these forces must eventually cancel out or converge to an optimal equilibrium. This approach prioritizes empirical fidelity to the complex, non-equilibrium nature of capitalist development over theoretical closure.

6 Decomposition Analysis

To distinguish between technological displacement and structural change, we complement the panel regressions with a shift-share decomposition of the aggregate labor share change:

$$\Delta LS_t^{\text{agg}} = \underbrace{\sum_s \theta_{s,t-1} \Delta LS_{s,t}}_{\text{Within Effect}} + \underbrace{\sum_s LS_{s,t-1} \Delta \theta_{s,t}}_{\text{Between Effect}} + \underbrace{\sum_s \Delta LS_{s,t} \Delta \theta_{s,t}}_{\text{Interaction}}, \tag{6}$$

where $\theta_{s,t}$ is the value-added share of sector s. This quantifies how much of the aggregate decline is due to falling shares within industries versus a reallocation of activity towards low-labor-share sectors (e.g., from manufacturing to digital platforms).

Addressing Innovation in the Service Sector

A critical challenge in measuring technological exposure for the service sector is the well-documented bias of patent data towards manufacturing and hardware. Service firms often innovate through non-patentable channels—such as proprietary software, algorithms protected as trade secrets, organizational capital, and new business models—leading to a potential underestimation of their technological intensity if relying solely on patent counts. Our design mitigates this measurement error through the *sector-of-use* attribution logic. By constructing exposure based on predetermined technological affinities ($\pi_{s\tau}$) rather than patent ownership, we capture the adoption of breakthrough technologies by service industries even when these technologies are developed by external suppliers (e.g., a bank adopting AI algorithms developed by a tech firm, or a retailer implementing automated logistics systems). The ALP concordance explicitly links technology classes relevant to services to their using industries, ensuring that service sectors receive appropriate exposure scores based on their *absorption* of radical technologies. To further address innovations that leave no patent trace whatsoever, we complement our baseline analysis for the period 1995–2025 with industry-level data on intangible investment (software, databases, and organizational capital) from the EUKLEMS & INTANProd database. This allows us to verify that our results are not driven by an omission of non-patented digital transformation in the service economy.

Optional Extension: The Green Transition and Labor Share

As a potential extension of the baseline analysis, this study can be expanded to disentangle the specific impact of the *green transition* from that of general-purpose breakthrough technologies. While our primary focus remains on radical innovations across all fields, the same Shift-Share

Local Projection design can be applied to a dedicated measure of **Green Breakthrough Exposure**.

To implement this, we would restrict the universe of patents to those classified under the *IPC Green Inventory* (WIPO) and the *EPO Y02* category (climate change mitigation technologies). Within this subset, we would construct a *Green Breakthrough Index* analogous to our baseline measure: each green patent would be weighted by its truncation-adjusted forward citations and technological originality (following Kelly et al., 2021), thereby isolating *radical* green innovations from incremental environmental adjustments. The exposure variable would then be reconstructed as:

$$\text{GreenExposure}_{s,t} = \sum_{p \in \text{Green}} \pi_{s\tau}^{\text{fixed}} \cdot B_p^{\text{green}} \cdot D(t - t_p), \tag{7}$$

using the same predetermined sector-of-use shares ($\pi_{s\tau}$) to ensure comparability with the baseline results.

This extension would allow us to test whether green technologies exhibit a different dynamic effect on the labor share compared to digital or AI-driven breakthroughs. Given that green innovations often entail significant capital deepening (e.g., renewable energy infrastructure) but may also be more labor-intensive in installation and maintenance phases, we hypothesize a potentially distinct impulse response profile. Including this analysis would provide policy-relevant insights into whether the ecological transition acts as a mitigating or amplifying force for the decline in the labor share, complementing the primary investigation into automation and digitalization.

7 Expected Contributions

This thesis aims to contribute to the literature in three ways:

1. **Methodological:** By applying Panel Local Projections to the study of technology and factor shares, we provide a more nuanced view of the adjustment dynamics than static panel models allow, while integrating an evolutionary perspective on identification.

2. **Empirical:** We offer new evidence on the heterogeneity of technological impacts across different historical waves, challenging the notion of a constant elasticity of substitution between capital and labor.

3. **Policy:** By disentangling within- and between-sector effects, we clarify whether the decline in the labor share is a result of pervasive automation or sectoral reallocation, informing targeted policy responses regarding taxation, education, and competition.

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